DESIGN AND DEVELOPMENT OF A WEB-BASED INVENTORY AND SALES MANAGEMENT SYSTEM FOR ELKIM CHEMISTRY

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1 INTRODUCTION

1.1 Background

Globally, the use of Information and Communication Technology (ICT) has greatly impacted business operations in various industries that require high accuracy, accountability, and timely decision-making processes, like the pharmaceutical industry (Ma et al., 2022). For instance, modern inventory and sales management systems have enabled various organizations to keep track of their stock in real-time, record sales, generate reports, and reduce losses arising from human error and fraud. Research shows that inaccurate inventory data, stock shortages, and insufficient real-time visibility are associated problems caused by manual inventory systems, which lead to poor operational efficiency and profits. Therefore, many SMEs across the globe are now widely utilizing a centralized, web-based inventory system for the purpose of integrating their operations, increasing transparency.

However, in many developing countries, Kenya included, a good number of small and medium pharmaceutical enterprises still depend on manual methods of inventory management, such as keeping handwritten records and spreadsheets. Despite having spreadsheets as an effective method that provides basic record-keeping functionality, they do not hold a feature for automation, real-time synchronization across multiple branches operating as one enterprise, or secure role-based access control where, an employee is entitled to access their branch's data only and not for all branches, unless authorized, and that the admin is able to view the data for all branches in one platform. The absence of these important dimensions within many Kenyan SMEs, and more so pharmaceutical outlets, exposes them to errors, manipulation of prices, reporting, and poor performance monitoring.

In densely populated areas of Kenya, such as urban settings, for example, Embakasi South and Embakasi East constituencies in Nairobi County, many small pharmaceutical outlets operate in conventional methods, perhaps without fully automated systems for inventory and sales management. In this context, the conventional methods used, such as manual recording on paper files and on spreadsheets, heighten the chances for pricing inconsistencies, where an employee may sell drugs at increased prices to customers who are unaware, unrecorded sales, stock discrepancies, and delayed identification of low-stock items. Since many outlets are managed by their owners, many times, the issue of low stock is very consistent because even if the employees notice it takes a lot of time as they communicate with the owner to arrange for restocking, which consumes a lot of time. Additionally, the business owners struggle to monitor performance by each branch due to the lack of a centralized reporting mechanism, where they rely on the information they receive from their employees when they make a visit to each branch or communicate through phones. This is complex because the owners do not have sufficient time to visit each branch regularly.

From this viewpoint, Elkim Chemistry, a pharmaceutical business outlet owned by Mr. John (fictitious name due to privacy) operates five branches within Nairobi County, with the main branch situated in Embakasi Pipeline, and other two branches also operating in Pipeline but on different areas of Pipeline, while the fourth branch operates in Mukuru Kwa Njenga area, with the fifth branch operating in Tassia in Embakasi East constituency. Elkim encounters operational challenges close to the ones underlined above, whereby it maintains its inventory and sales records through manual methods involving file records and spreadsheets for calculations. Thus, the outlet lacks a centralized system that integrates operations across all branches, such that John can perform business evaluations and track sales and performance for each branch. As

a result of this gap, Elkim faces multiple consequences, including a series of inaccuracies based on stock monitoring, auditing, branch performance comparison, and decision-making, which are time-consuming. In addition, the lack of automation facilitates the absence of low-stock alerts, which can show the drugs that are nearly running out of stock, and this results in unexpected stockouts, forcing loyal customers and one-time customers to seek services elsewhere.

1.2 Introduction: Research Area

The study can be classified as an Information Systems and Software Engineering discipline, with more specific areas of interest being web-based enterprise systems and inventory management solutions. Information systems for enterprises have been developed to integrate business processes into one system for the purpose of real-time monitoring and control, as well as enhanced decision-making processes (Nisrina et al., 2024; Killedar et al., 2024).

Across the globe, there is a shift from separate and manual systems towards centralized web-based systems that offer dashboards, reporting, performance analytics, and user management. These systems help increase transparency, reduce fraud, and increase accountability. In the pharmaceutical retail sector, digital inventory management is of critical importance for ensuring availability, avoiding expiration of medicines, and for effective management of stocks.

Despite these developments, many SMEs in Kenya, especially small pharmaceutical retailers, have not fully embraced and implemented automated enterprise systems. This research, therefore, seeks to bridge the gap between global ICT developments and their actual application in local multi-branch pharmaceutical enterprises. This research adds to the increasing pool of knowledge on how ICT applications can facilitate business operations and governance for small and medium businesses.

1.3 Problem Statement

Elkim Chemistry and similar SMEs, even those operating in other sectors such as retail and hotels in Embakasi South and Embakasi East constituencies of Nairobi county, which form the most densely populated areas in Nairobi, face significant operational inefficiencies due to the reliance on manual methods of inventory and sales management, perhaps including paper or file work and spreadsheets, record keeping, and calculations. The outlet lacks a centralized framework for managing its five branches operating in different locations within the two sub-counties, and this has created accountability gaps, data inconsistencies, and challenges associated with performance monitoring.

The dimensions involving manual recording of sales and inventory intensify the potential for human error by employees working in these branches, whereby they often forget to record some daily transactions, enter incorrect quantities, or often miscalculate the total daily or weekly sales. Since all the branches operate 24 hours a day, with day and night shifts, employees on each shift often blame each other for any issues emerging due to a lack of monitoring on when an issue occurred and who was involved. These errors affect the stock records directly, as well as impacting the generation and storage of clean financial reports and records. Additionally, these manual systems facilitate unethical contact by the staff since an employee may sell drugs at inflated prices to gain personal benefit. On some occasions, employees may plan to avoid recording the sales in a file or spreadsheet and retain the cash, and this may affect others who may be scrutinized to explain the whereabouts of the cash sales and even pay for it when noticed. Since Elkim lacks a centralized automated system to track sales patterns and data in real time, such discrepancies remain undetected until a physical stock count is conducted.

Additionally, stock mismanagement presents a serious issue in all branches of Elkim. On many occasions, drugs are sold without proper documentation, and this results in unexplained shortages. In cases when these issues are noticed, employees are often unfairly accused or penalized for loss of cash sales that have been stolen by their colleagues. Thus, this affects staff morale (among those accused unfairly) and trust between management and staff. Basically, a lack of transparency can facilitate the development of internal conflicts and, hence, diminish overall productivity (Almahraj & Hejazi, 2025). The lack of a system also leads to the absence of performance evaluations for each branch, which can hinder effective decision-making. Indeed, the daily sales cannot be compared in detail, as only the total daily sales from each branch are submitted to the owner in raw figures after each shift. This is therefore linked to limiting data-driven decision-making about timely procurement, pricing adjustments, and branch supervision. Additionally, the staff struggles to identify which drugs are nearing expiry dates and plan for their use in a timely manner to avoid having them expire and get disposed of.

Auditing processes are also complex and time-consuming. Determining the number of drugs under each category, verifying sales totals, and reconciling stock discrepancies require manual effort. The absence of automated low-stock alerts means that essential drugs may run out unexpectedly. When customers find medicines unavailable, they often seek alternatives from competing pharmacies, leading to loss of revenue and potential long-term customers. The magnitude of these issues impacts financial performance, customer satisfaction, employee relations, and the sustainability of the business. There, therefore, exists a need for a central, automated system to improve transparency and accountability, as well as operational efficiency in all branches of the business.

1.4 Proposed Solution

The research project seeks to design and develop a centralized web-based inventory and sales management system for Elkim Chemistry that will serve to automate all the necessary inventory dimensions, including inventory tracking, sales recording, payment monitoring, and conducting evaluations for branch performance. The solution is aimed at replacing the manual file/paper and spreadsheet-based record keeping with an integrated platform accessible to all branches through a secure role-based access control. The system will function to centralize Elkim's operations across all its five branches into a unified database interface, allowing the admin (the business owner) to monitor stock levels, sales performance, payment modes (either cash or M-PESA), drugs nearing out of stock, drugs nearing expiry dates, and all other branch activities in real time through a dashboard interface.

The components of research specific to this project will involve analyzing the best practices in modern inventory management systems, pinpointing the functional requirements unique to Elkim Chemistry, and evaluating the effectiveness of the developed system in enhancing operational efficiency. The solution will integrate automated stock updates after each sale, low-stock alert mechanisms as per the predefined thresholds, expiry dates' alert mechanisms based on the predefined dates for alerts, and branch-level performance reporting. By aligning with global enterprise system models that emphasize transparency, centralization, and data-driven decision-making, this research will demonstrate how SMEs in Nairobi can leverage ICT solutions to overcome operational inefficiencies and strengthen internal controls.

1.5 Objectives

General Objective: To design, develop, and evaluate a centralized web-based inventory and sales management system that enhances efficiency, transparency, and branch coordination at Elkim Chemistry within a one-and-a-half-month period.

Specific Objectives

1. To analyze the existing inventory and sales management processes at Elkim Chemistry and identify operational weaknesses within the first week.
2. To design a centralized system architecture integrating all five branches with role-based access control by the second week.
3. To develop and implement a web-based inventory and sales management system that automates stock updates, sales recording, and payment tracking by the fourth week.
4. To incorporate a low-stock alert mechanism that notifies management when drug quantities fall below predefined thresholds and an alert mechanism that notifies when a drug is nearing expiry date.
5. To evaluate the system's effectiveness in reducing discrepancies, improving reporting accuracy, and enhancing branch performance monitoring by the eighth month.
6. To document findings and recommend best practices for similar SMEs seeking digital transformation.

1.6 Justification

The importance of this research is seen in relation to Elkim Chemistry, other SMEs in Nairobi, and Information Systems in general. To Elkim Chemistry, the system is important because it will help reduce inaccuracies in inventory and sales records, minimize losses from unrecorded transactions, and increase accountability. The centralized dashboard will enable the

owner to monitor branch performance in real time, detect irregularities early, and make informed procurement and pricing decisions. Other small and medium enterprises that use similar manual solutions can look to this study to digitize their entire operations. This study will show how using a web-based solution can increase transparency, improve customer satisfaction, and enhance operational efficiency for any enterprise facing limited resources.

Academically, the research contributes to the application of software engineering principles in solving real-world business problems. It bridges theoretical ICT knowledge with practical implementation in a local context.

1.7 Proposed Research & System Methodologies

This study will be based on a structured methodology that will involve a sequence of phases starting with gathering the requirements, performing comprehensive system analysis and physical design, developing the system and implementing it, testing the developed system, and evaluating the system functionality, which will be based on determining if the system has addressed all the user requirements. In this context, data that will lead to gathering the requirements will be collected through conducting an interview with the business owner and the staff, directly observing the existing processes, and analyzing the current manual and spreadsheet records systems. In the system development phase, the Software Development Lifecycle (SDLC) iterative model will be employed. This model will allow for progressive refinement of system components based on the feedback. The system modeling will be informed by the principles of Object-Oriented Analysis and Design (OOAD), which will guarantee modularity, scalability, and the maintainability of the system. These methodology approaches will be appropriate because they will provide systematic procedures for translating the user requirements into a reliable and functional system.

1.8 Scope

Geographically, this study will focus on the five branches of Elkim Chemistry located in Embakasi South and Embakasi East constituencies within Nairobi County. The target users include the business owner (Administrator) and branch staff responsible for inventory and sales operations. Functionally, the system will cover inventory management, sales recording, payment mode tracking (Cash and M-PESA), low-stock alerts, expiry dates alerts, and branch performance reporting. The study will not include patient medical records, prescription management, supplier integration systems, or clinical decision-support tools. It will strictly focus on inventory and sales management operations within the selected enterprise.

2 CHAPTER TWO: PROBLEM ANALYSIS

2.1 Description of the Problem Domain

2.1.1 Description of the Theoretical Domain

Inventory and sales management systems fall under the broader field of Information Systems within business process automation, where technology is used to improve efficiency, accuracy, and decision-making in organizational operations. Globally, organizations adopt Management Information Systems (MIS) and Enterprise Resource Planning (ERP) systems to integrate business functions and ensure seamless flow of information. In the pharmaceutical sector, inventory management is particularly critical due to the sensitive nature of drugs, the need to track expiry dates, maintain consistent pricing, and respond to fluctuating demand. Effective systems ensure proper stock control, accountability, and availability of drugs, which directly affects customer satisfaction and business performance.

The theoretical foundation of such systems is based on concepts such as centralized database management, real-time data processing, role-based access control, and decision-support

systems. Web-based systems operate on a client-server architecture, where all data is stored in a centralized database and accessed through secure authentication mechanisms. This approach enhances transparency, reduces redundancy, and prevents manipulation of records. Additionally, principles of system analysis and design, data integrity, and human-computer interaction are applied to ensure that systems are efficient, reliable, and user-friendly. However, in many developing contexts, especially among small and medium enterprises, these theoretical advantages are not fully realized due to continued reliance on manual or semi-automated systems such as spreadsheets, which lack automation and real-time capabilities.

2.2 Identification of Target Population

The target population for this study consists primarily of stakeholders within Elkim Chemistry, a pharmaceutical business operating five branches within Nairobi County. This includes the business owner, who acts as the system administrator and requires centralized control and visibility over operations across all branches, as well as branch staff who are responsible for recording sales transactions and managing inventory at their respective locations. These users interact directly with the current manual system and are therefore most affected by its limitations.

In addition to the internal stakeholders, the study also indirectly considers customers who depend on the availability of drugs and fair pricing. Furthermore, the findings of this study are applicable to other small and medium pharmaceutical enterprises operating in areas such as Embakasi South and Embakasi East, where similar challenges exist. Many of these businesses rely on manual systems, making the study relevant beyond the immediate case of Elkim Chemistry.

2.3 Techniques for Data Collection

To obtain a clear understanding of the existing system and its associated challenges, qualitative data collection methods were employed. Interviews were conducted with the business owner to gather detailed information about the operational difficulties experienced, including stock discrepancies, sales tracking issues, and lack of centralized control. These discussions provided valuable insights into the real-world challenges faced in managing multiple branches using manual systems.

In addition, direct observation was carried out within the business premises to examine how inventory and sales are currently recorded. This involved analyzing the use of spreadsheets and manual records, which revealed inefficiencies such as delayed updates, lack of standardization, and vulnerability to manipulation. Document review was also conducted by examining existing records, which helped identify inconsistencies, absence of automated alerts, and limitations in reporting capabilities. These data collection techniques ensured a comprehensive understanding of the problem domain.

2.4 Review of Existing and Similar Systems

Existing systems used in inventory and sales management vary from advanced enterprise solutions to simple spreadsheet-based tools. Globally, large pharmaceutical organizations utilize Enterprise Resource Planning systems and specialized pharmacy management systems that provide centralized control of inventory and sales operations. These systems automate stock updates, enforce pricing consistency, generate real-time reports, and provide alerts when stock levels fall below predefined thresholds. By integrating all branches into a single platform, such systems enhance transparency, accountability, and efficiency in operations.

However, these advanced systems often present significant challenges when applied to small and medium enterprises. They are typically expensive to acquire and maintain, require specialized technical expertise, and may include complex features that are not necessary for smaller businesses. As a result, many SMEs opt for simpler tools such as spreadsheets to manage their operations. While spreadsheets allow basic data entry and calculations, they lack essential features such as real-time synchronization, centralized access, role-based security, and automated alerts. This leads to several operational issues, including data inconsistencies, difficulty in auditing, and the inability to generate consolidated reports across branches.

In the case of Elkim Chemistry and similar businesses, reliance on spreadsheets contributes to problems such as stock discrepancies, unrecorded sales, and inconsistent pricing. Employees may manipulate records or fail to update them accurately, leading to financial losses and conflicts within the organization. Additionally, the absence of a centralized system makes it difficult to monitor branch performance or detect irregularities in real time. Although existing systems provide partial solutions, they either lack the necessary functionality or are not suitable for the operational and financial context of small-scale businesses.

2.5 Feasibility Study

A feasibility study was conducted to assess the viability of implementing a web-based inventory and sales management system for Elkim Chemistry. Technical feasibility considers whether the necessary technology and expertise are available to develop and deploy the system. The proposed system will utilize PHP and SQL Server, which are reliable and widely used technologies for web application development. The developer has the required knowledge and skills to implement the system, and the necessary hardware resources, including a personal computer and internet connectivity, are available. Therefore, the project is technically feasible.

Economic feasibility evaluates the cost implications of the project in relation to its expected benefits. Since the system is being developed as part of an academic project, development costs are minimal and limited to essential resources such as domain registration, hardware upgrades, and backup storage. Compared to commercial systems that require significant investment, the proposed solution is cost-effective and sustainable for the business. Operational feasibility examines whether the system will be accepted and effectively used within the organization. The system is designed to align with existing workflows while improving efficiency, and the business owner has shown willingness to adopt it. Additionally, the user interface will be simple and user-friendly, ensuring that staff can easily transition from manual processes to the automated system.

Schedule feasibility considers whether the project can be completed within the available timeframe. Given that the project runs from early February to mid-March and focuses only on core functionalities such as inventory and sales management, it is achievable within the semester period. By limiting the scope to essential features and following a structured development plan, the project can be successfully completed within the given timeline.

3 CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Identification of SAD Methodology

This project adopts the Structured Systems Analysis and Design Method (SSADM) as the approach for analyzing and designing the system. SSADM is suitable for this system because it focuses on structured processes, data flow, and logical organization of system components. The system being developed is data-driven and involves clearly defined operations such as product management, customer handling, supplier management, and sales processing. SSADM provides a systematic way of understanding how data moves within the system and how different

processes interact with the database. It also allows separation between logical design and physical implementation, making it easier to develop a well-organized and maintainable system.

3.2 Analysis of Collected Data Using Selected Method

The analysis of the collected data shows that the system is designed to manage inventory and sales operations through several key data entities. These include users, products, categories, customers, suppliers, and sales. Users interact with the system and perform operations such as adding products, recording sales, and managing customer and supplier information. The products entity is central to the system and contains details such as product name, generic name, category, supplier, original price, selling price, quantity, as well as arrival and expiry dates. The inclusion of arrival and expiry dates enables the system to track product lifecycle and monitor stock validity.

Categories are used to group products for easier organization, while customers and suppliers store information about individuals or organizations involved in transactions. The sales entity captures transaction details including the product sold, customer, quantity, payment, and date of sale. Data flows from the user into the system through input forms, is processed by the system, and is then stored in the respective database tables. The system also retrieves stored data to display information such as product listings, sales records, and reports.

3.3 System Requirement Specification

The system requirements define what the system is expected to do and how it should perform. The functional requirements of the system include the ability to allow users to log in and log out, manage user accounts, and perform operations such as adding, updating, deleting, and viewing products, categories, customers, and suppliers. The system should also record sales transactions, calculate profit or loss based on product prices, update product quantity after each

sale, and generate reports. In addition, the system must record product arrival and expiry dates and support monitoring of expired or near-expiry products.

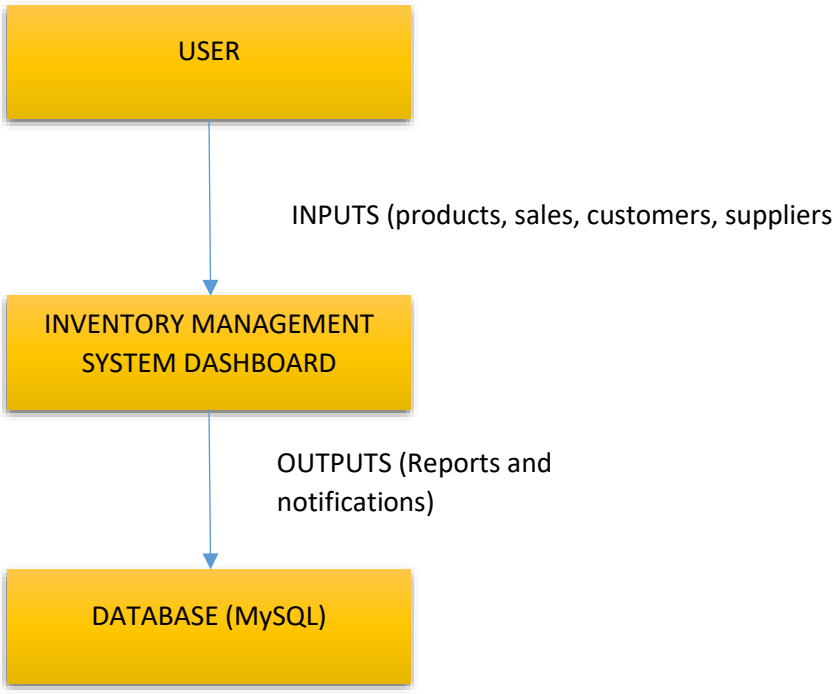
The non-functional requirements focus on system quality attributes. The system should ensure security by requiring user authentication and protecting user data. It should provide good performance by responding quickly to user actions and processing data efficiently. Usability is also important, and the system should have a simple and user-friendly interface. Reliability is required to ensure that data is stored correctly and errors are handled appropriately. The system should also be maintainable so that it can be updated and improved easily when needed.

3.4 Design of the System According to Requirements Specified

3.4.1 Logical Design

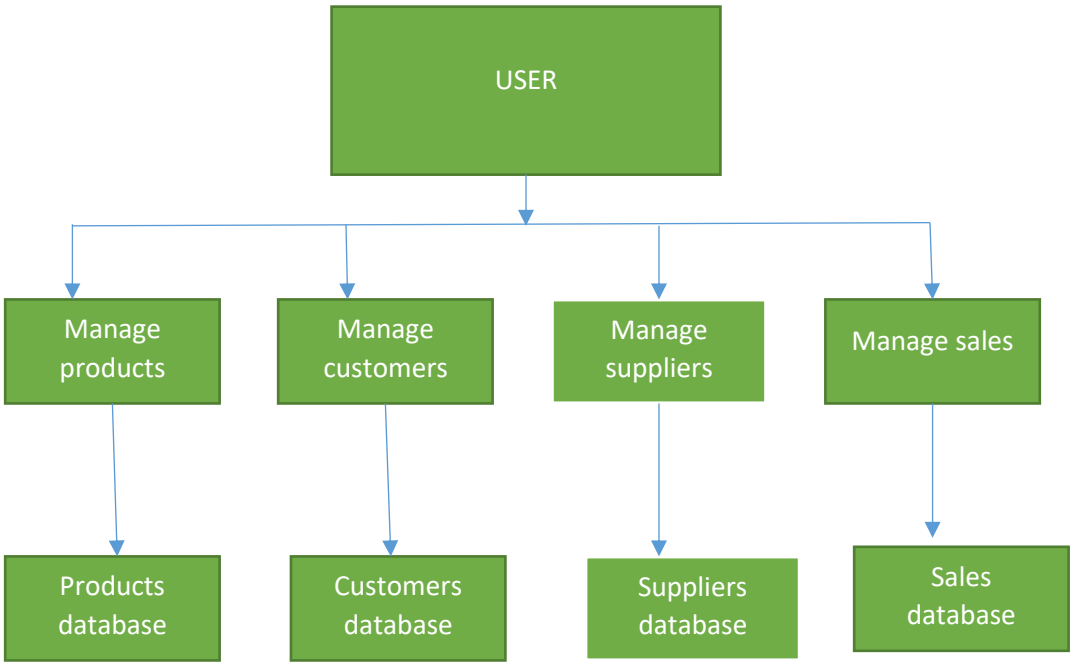
The logical design represents how data flows within the system without focusing on physical implementation. It shows how users interact with the system and how data moves between processes and data stores.

3.4.1.1 Context Diagram (Level 0 DFD)



The context diagram shows the system as a single process interacting with the external entity, which is the user. The user provides input such as product details, customer information, and sales data. The system processes this input and stores it in the database, and then provides output such as reports and notifications back to the user.

3.4.1.2 Level 1 Data Flow Diagram (DFD)



The Level 1 DFD breaks the system into major processes. These include managing products, customers, suppliers, and sales. Each process receives input from the user, processes the data, and stores it in the respective database table. The product management process includes handling important fields such as arrival date and expiry date, which are essential for inventory tracking.

3.5 Physical Design

The physical design describes how the system is implemented in terms of inputs, outputs, storage, processing, and control.

Input Requirements: The system accepts input through various forms. These include product forms, customer forms, supplier forms, and sales forms. The product form includes fields such as product name, generic name, category, supplier, arrival date, expiry date, original

price, selling price, and quantity. These inputs ensure that all necessary product details are captured for proper inventory management.

Output Requirements: The system generates outputs such as product lists, sales reports, customer records, and inventory summaries. It also provides notifications to inform users about successful or failed operations. Additionally, the system supports exporting data to formats such as Excel for reporting purposes.

Storage Requirements: The system uses a relational database to store data. The database consists of tables for users, products, categories, customers, suppliers, and sales. Each table stores structured data relevant to system operations. The product table includes fields for arrival and expiry dates, which help in tracking the product lifecycle. Although these fields are currently stored as text, it is recommended that they be stored using proper date data types for improved data integrity and validation.

Processing Requirements: The system performs several processing activities including validation of user input, calculation of profit or loss, updating product quantities after sales, and handling communication between the user interface and the server. It also processes date-related information such as arrival and expiry dates to ensure accurate inventory tracking.

System Control and Backup: The system includes control mechanisms such as user authentication, input validation, and notification handling to ensure proper system operation. Backup and recovery can be achieved through regular database exports, which help prevent data loss and ensure system reliability.

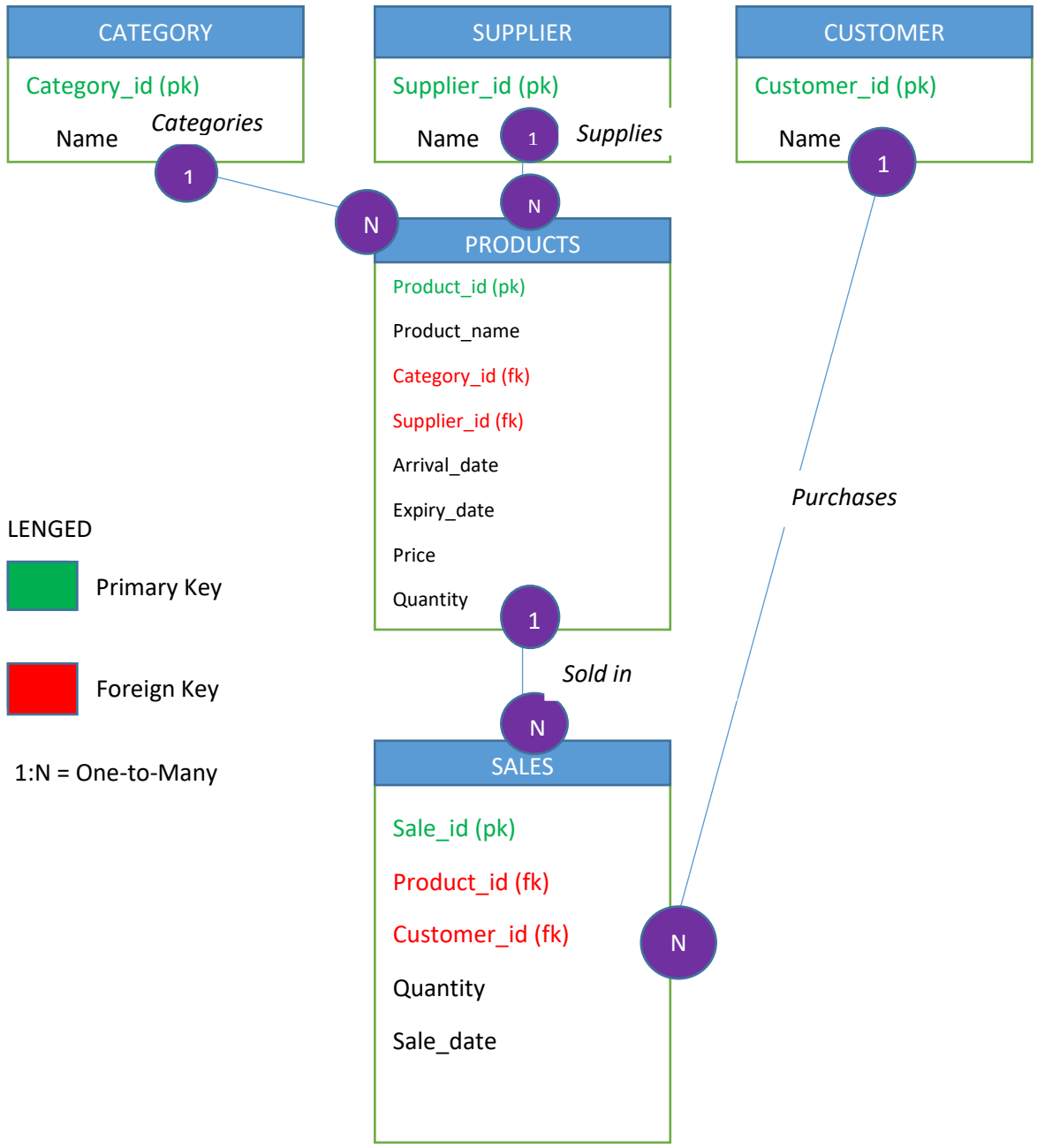
3.6 Detailed Design

3.6.1 User Interface Design

The user interface is designed to be simple and user-friendly, allowing users to interact with the system easily. It includes dashboards and forms for managing products, customers, suppliers, and sales. The product interface allows users to input important details such as arrival and expiry dates, ensuring complete tracking of product information. After login into the system, dashboard is displayed first, showing numerous things that point to the functionalities of the system. For instance, it shows a column in the left that has the menu which lists important features like creating product, which is basically adding a product into the system such that they are managed through the system instead of through the manual system. The menu also lists reports which presents all the reports about sales, stock value, customers, top selling products, payments, stock alerts, payment methods etc. The menu contains the users, including the admin and the recognized users who are permitted to use the system. This section has the suppliers who supply drugs to Elkim, customers, and the system users who are basically the employees. The settings are also in the dashboard, where system settings, backup, system update, and restart are. Then there is a logout form, which permits logging out of the system. The dashboard is user-friendly, such that when users interact with the system, they just need to click the function they need to perform and fill in the details. For instance, drug categories such as antibiotics, antifungals etc are created only through clicking “add category” and thereafter, they can add products into such categories easily as they need to click “add products” and a pop-up window-like form appears on the screen with fields open for filling the details of a product and after filling these details, they click save and product is added into the system and classified into its category.

3.7 Data Design

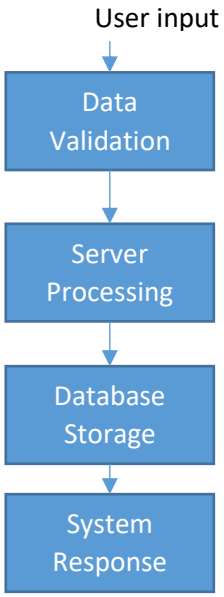
The data design describes how data is structured within the system. The system uses multiple tables, each representing a specific entity such as users, products, categories, customers, suppliers, and sales. Each table includes a primary key for identification and contains relevant attributes. The design recommends the use of appropriate data types, such as DATE, for storing arrival and expiry dates, as well as the use of relationships between tables to improve consistency and reduce redundancy. Below is the Entity Relation Diagram showing the entity relations in the database.


CATEGORY
Category_id (pk)
Name
Categories
1
N
SUPPLIER
Supplier_id (pk)
Name
1
Supplies
N
CUSTOMER
Customer_id (pk)
Name
1
PRODUCTS
Product_id (pk)
Product_name
Category_id (fk)
Supplier_id (fk)
Arrival_date
Expiry_date
Price
Quantity
1
Sold in
N
Purchases
SALES
Sale_id (pk)
Product_id (fk)
Customer_id (fk)
Quantity
Sale_date
N
LENGED
Primary Key
Foreign Key
1:N = One-to-Many

From the ERD above, the Primary Key (PK) is the unique identifier for each record within a table, ensuring that every row can be differentiated from all others. This means that no records are sharing the same primary key value and also, it cannot be null or empty. In the database, category_id, supplier_id, and customer_id, as well as product_id and sales_id all function as primary keys, allowing the database to quickly locate specific records and maintain data integrity. On the other hand, Foreign Key (FK) serves as the link between related tables by referencing the primary key of another table. Foreign keys establish the relationships that make the relational database powerful and efficient. For instance, from the ERD, that shows Elkim's database, the PRODUCT table, the category_id field is the foreign key that points to the category_id primary key in the CATEGORY table, indicating which category each product belongs to. In the same way, the supplier_id in the PRODUCT table links to the SUPPLIER table while the product_id and customer_id in the SALES table connects the sales records to the specific products and customers. These foreign key relationships enforce referential integrity, meaning that the database prevents Elkim from creating a product with a non-existent category or recording a sale for a product that does not exist, ensuring that all connections between data remain valid and consistent.

3.8 Process Design

The process design explains how data flows through the system from input to storage and output.



The process begins when the user enters data into the system through input forms. The system validates the data to ensure correctness before sending it to the server. The server processes the data and stores it in the database. Once the operation is complete, the system sends a response back to the user in the form of a notification. This ensures that all data is properly handled and that users receive feedback on their actions. For instance, when a user clicks "add product" it opens a field where the details of the product such as product name, category, generic name, the supplier, quantity and date it is added and the date it expires are filled. The user clicks save and if one field is not filled, the system returns a notification to fill it. If the data is correct, it is validated and server processes it and stores it in the database and a notification is send "*success! Operation complete*." This means that the product has been added into the system successfully and its components are stored within the database.

4 CHAPTER FOUR: SYSTEM IMPLEMENTATION

4.1 Code Generation

The system was implemented as a web-based Inventory and Sales Management System using a combination of modern web technologies. The development followed the specifications defined in earlier chapters to ensure that all required functionalities were fully realized. The system uses PHP for server-side processing, JavaScript for client-side interaction, HTML for structuring web pages, and CSS for user interface design. The database was implemented using MySQL, accessed through the XAMPP server environment with phpMyAdmin used for database management. The code was developed using Sublime Text as the primary code editor.

4.1.1 Hardware Resources

The system was developed and tested using a standard personal computer equipped with sufficient processing power, memory (at least 4GB RAM), and storage. Input devices such as a keyboard and mouse were used for entering system data, while a display monitor provided the graphical interface for user interaction.

4.1.2 Software Resources

The software environment consisted of XAMPP, which provided Apache as the web server and MySQL as the database server. phpMyAdmin was used to create and manage database tables such as users, products, categories, suppliers, customers, and sales. A web browser was used to run and test the system.

The backend logic was implemented using PHP scripts such as main_server/server1, server2, usercrater, and loginserver. These scripts handle core operations including data insertion, updating, deletion, authentication, and retrieval. On the client side, JavaScript

functions were used to handle user interactions and asynchronous communication with the server using AJAX (XMLHttpRequest).

4.2 Actual Software Code Generation

The system functionalities were implemented according to the system requirements. Product management was implemented using JavaScript functions such as getnewproduct(), which collects product details including name, category, supplier, price, and quantity, and sends them to the server using a POST request. The PHP backend processes the request and stores the data in the MySQL database.

Customer and supplier management functionalities were implemented using functions such as getnewcustomer() and getnewsupplier(), which communicate with the server to store and update records. Category management is handled using getnewcategory().

User account creation and management were implemented using the createuser() function, which sends user details including name, email, password, and profile image to the server. Authentication is handled through the loginserver, where users log in using their email and password stored in the database. Only users approved by the administrator are allowed access, ensuring system security.

Sales functionality is implemented by allowing users to select products from the system. Once a product is selected, its details such as price are automatically retrieved, and the user inputs the quantity. The system then calculates the total sales value and profit in real time using JavaScript functions such as profiter(). The sales data is then sent to the backend and stored in the database.

The system also includes additional utility functions such as notifier() for displaying success or error messages, exportTableToExcel() for exporting data, and globaleditor() for retrieving existing records for editing. These functions improve usability and system efficiency.

4.3 Software System Integration

System integration involved combining the frontend interface, backend logic, and database into a unified working system. The frontend, developed using HTML and CSS, provides forms and dashboards through which users interact with the system. JavaScript enhances this interface by handling dynamic actions such as form submissions, calculations, and real-time updates.

AJAX (using XMLHttpRequest) is used to enable asynchronous communication between the client and server. This allows data to be sent and received without refreshing the page, improving system responsiveness. For example, when a user adds a product or records a sale, the data is sent to the server in the background, processed, and a response is returned instantly.

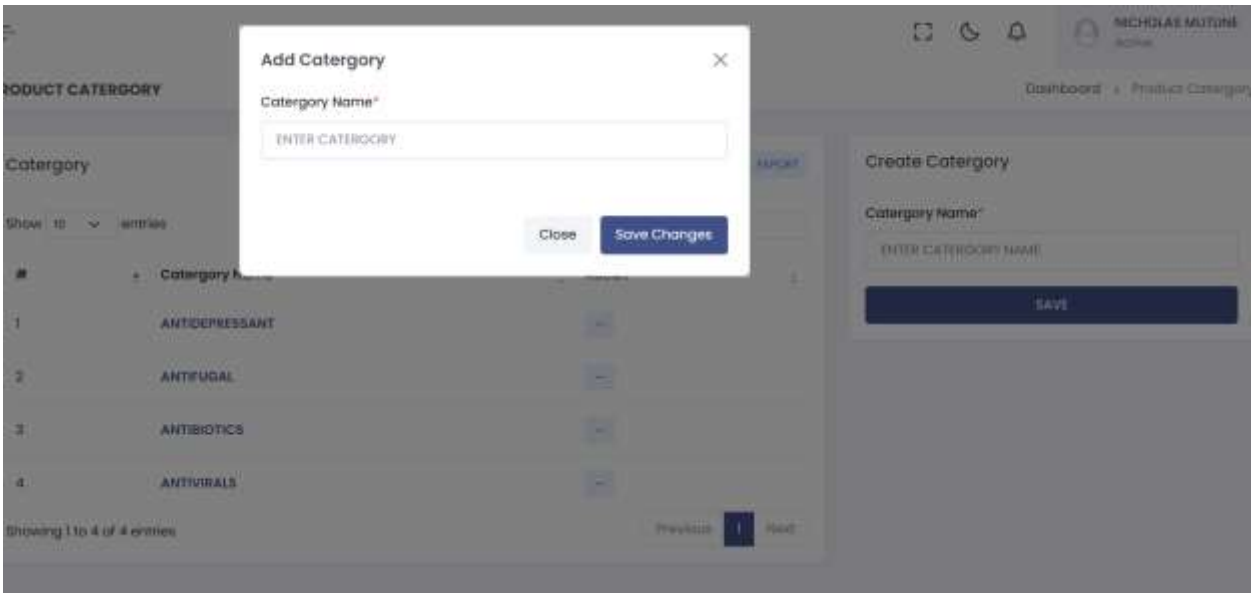
The PHP backend scripts process incoming requests and interact with the MySQL database. Data is stored in structured tables such as products, customers, suppliers, categories, users, and sales. The integration ensures that all components work together seamlessly, from data entry to storage and retrieval.

The system dashboard integrates various components by displaying key information such as daily sales, total sales trends, and top-selling products. Sales data is visualized using charts, including a line chart showing sales over seven days and a pie chart showing the most sold products based on quantity. These visualizations are generated dynamically based on data retrieved from the database.

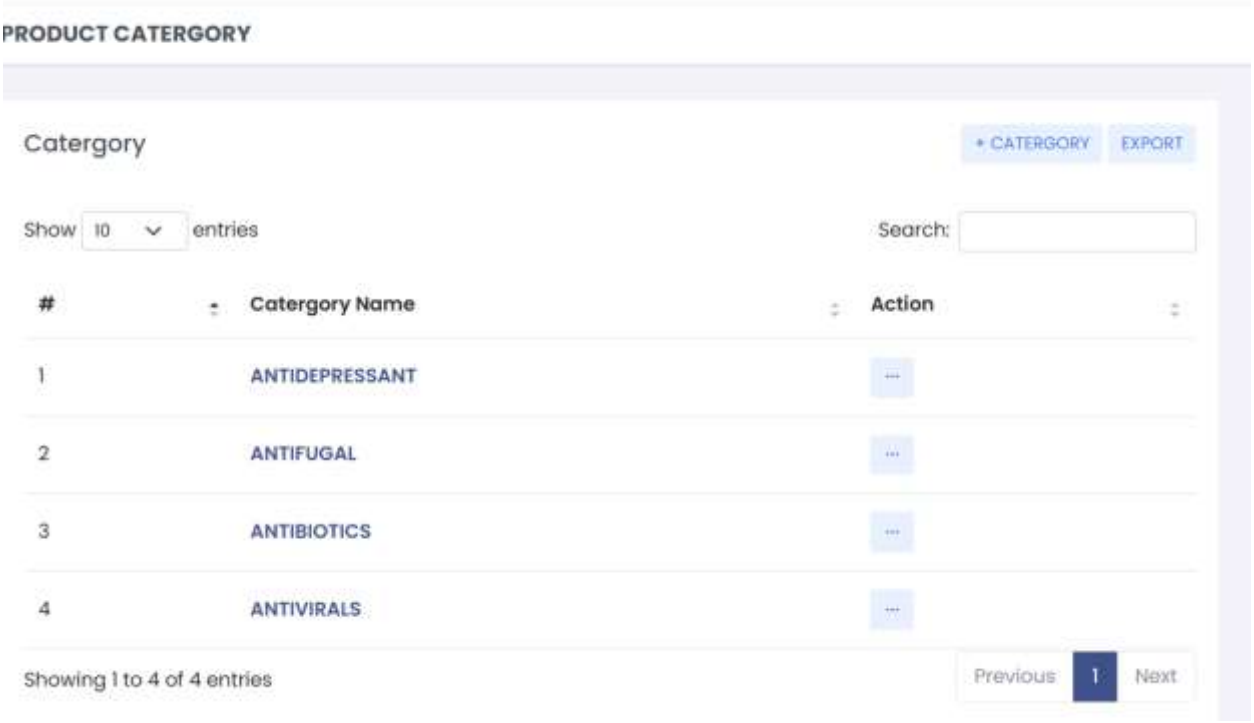
The overall system follows a client-server architecture, where the browser acts as the client and the PHP scripts on the XAMPP server act as the server. This architecture ensures efficient processing, modular design, and ease of maintenance.

4.4 Software System Testing

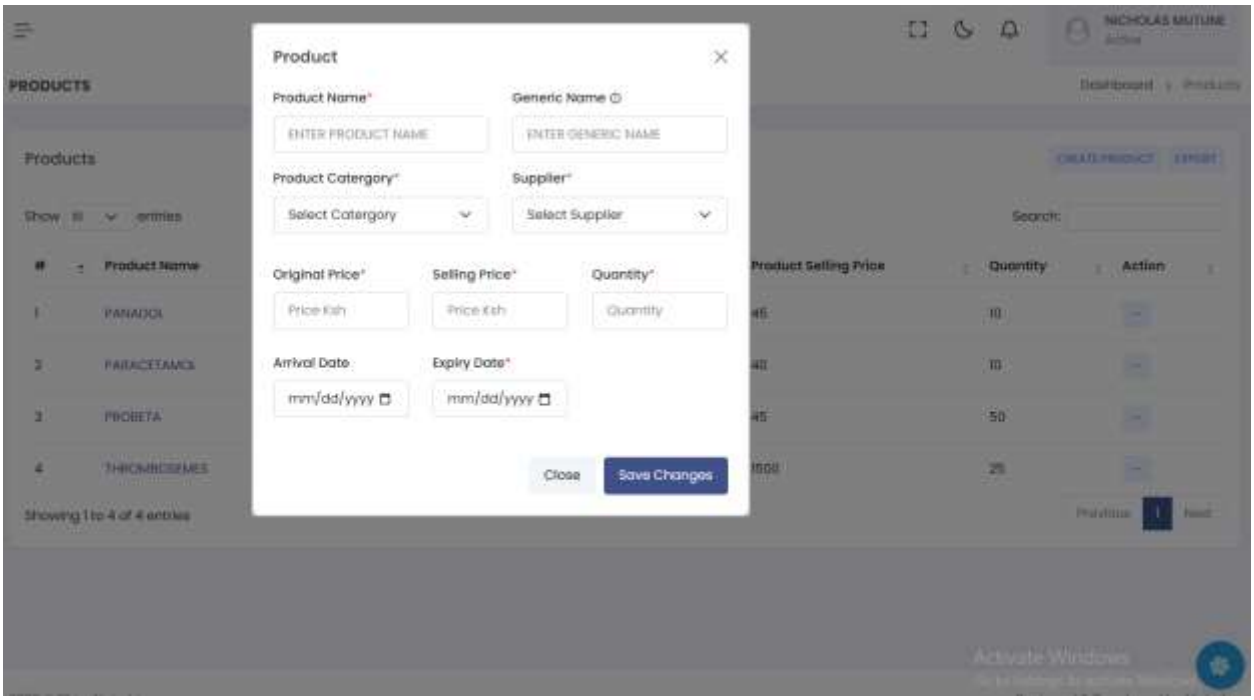
Software testing was conducted to verify that the system meets the specified requirements and functions correctly under different conditions. Testing involved executing system features, observing outputs, and comparing them with expected results. The following screenshots shows the working of the major functionalities of the system.



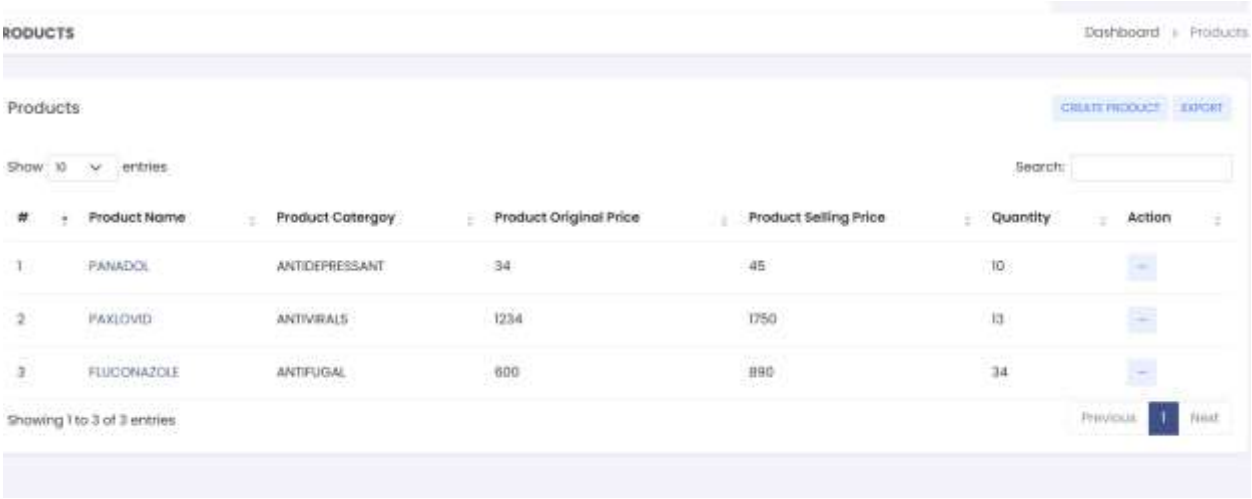
Results after adding product categories



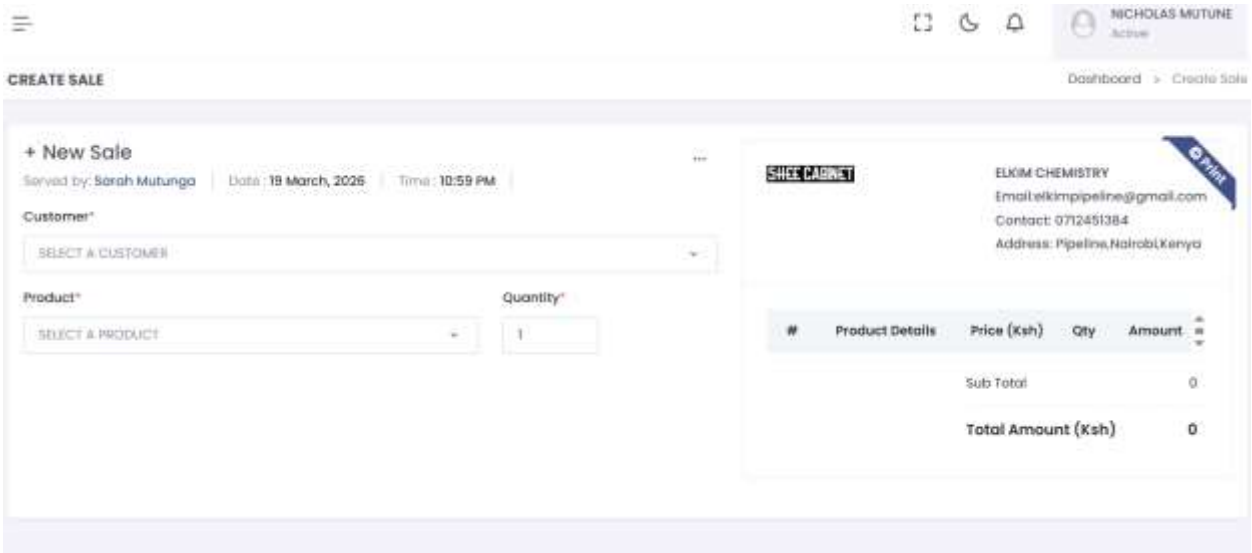
Adding products into the system is shown below



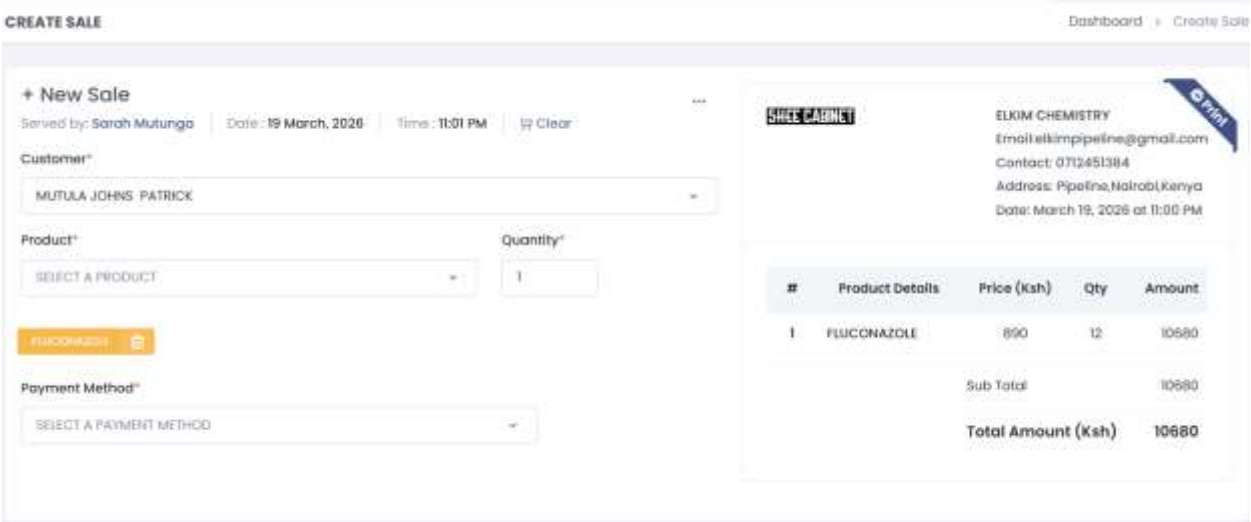
After adding products, they are displayed in the poducts section which is accessible in the dashboard



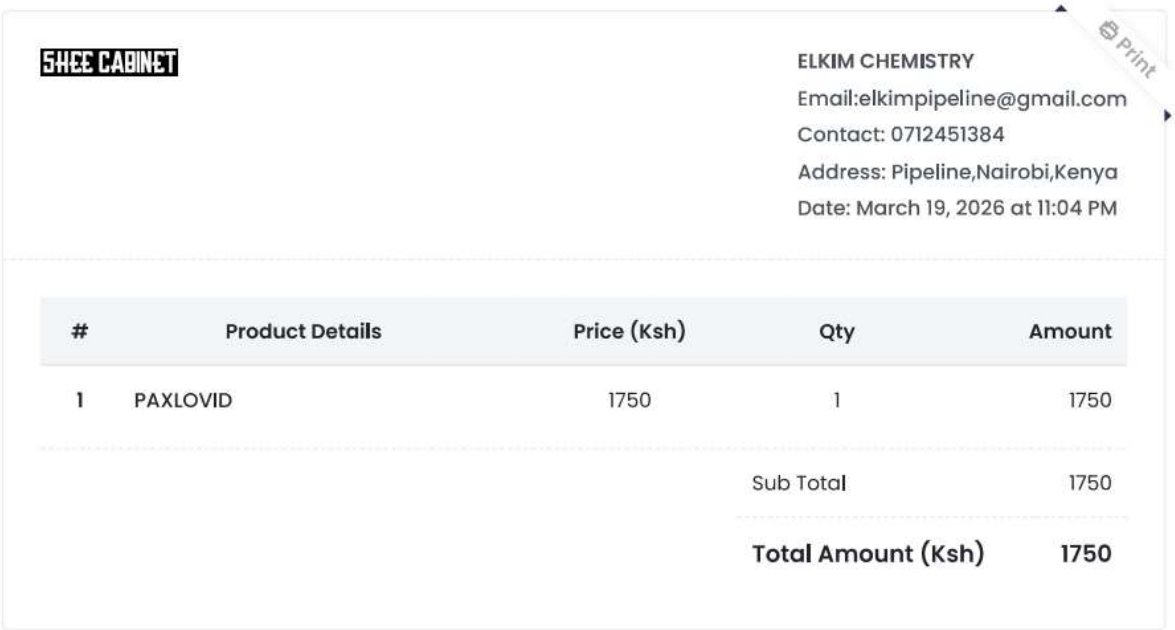
Below is a form that appears when a system user, an employee wants to sell a product within the system



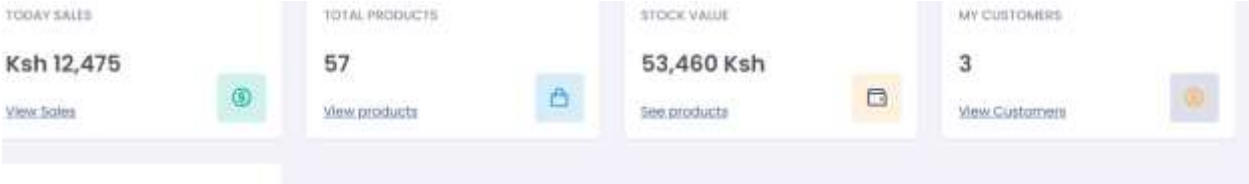
After inputting all the fields and saving, it appears like this, allowing the user to input the payment method used by a customer and save the sale, which is saved on the database and a report can be generated and stored within the permanent storage of a computer since it permits printing the receipt in PDF.



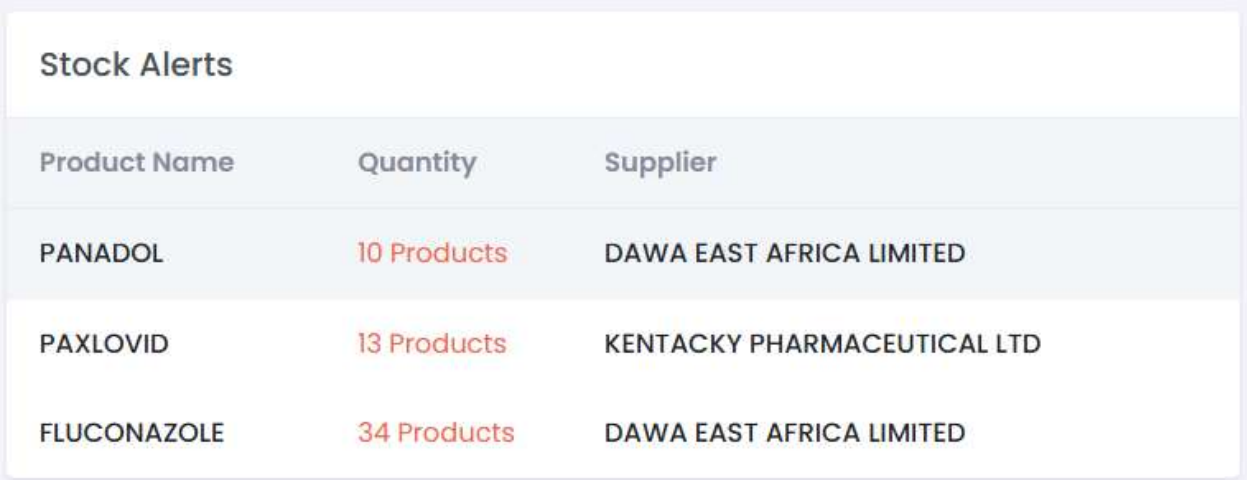
A receipt for each sale appears like this and can be downloaded and saved



Sales are recorded here



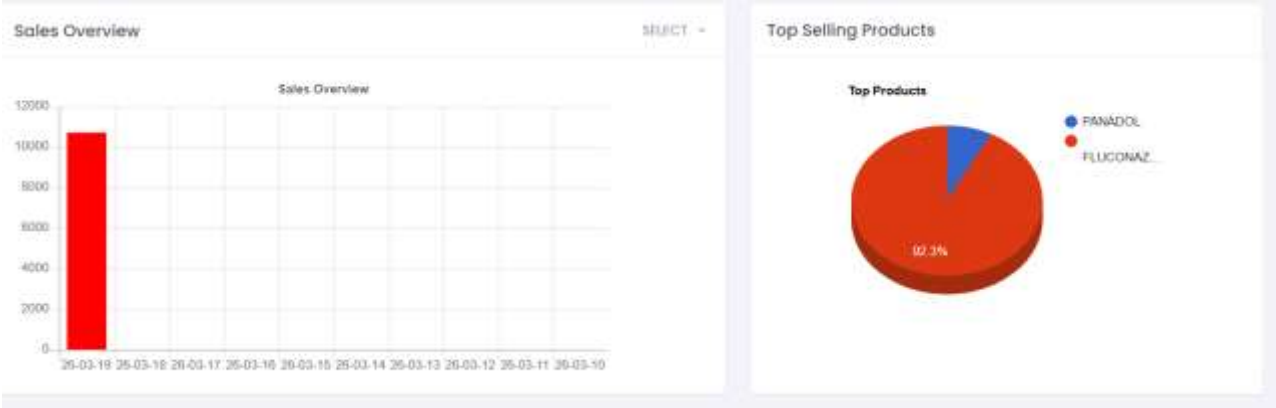
Stock alerts shows the remaining drugs as they are sold and reduces them as they are being sold and also adds and updates the correct number when products are added into the system

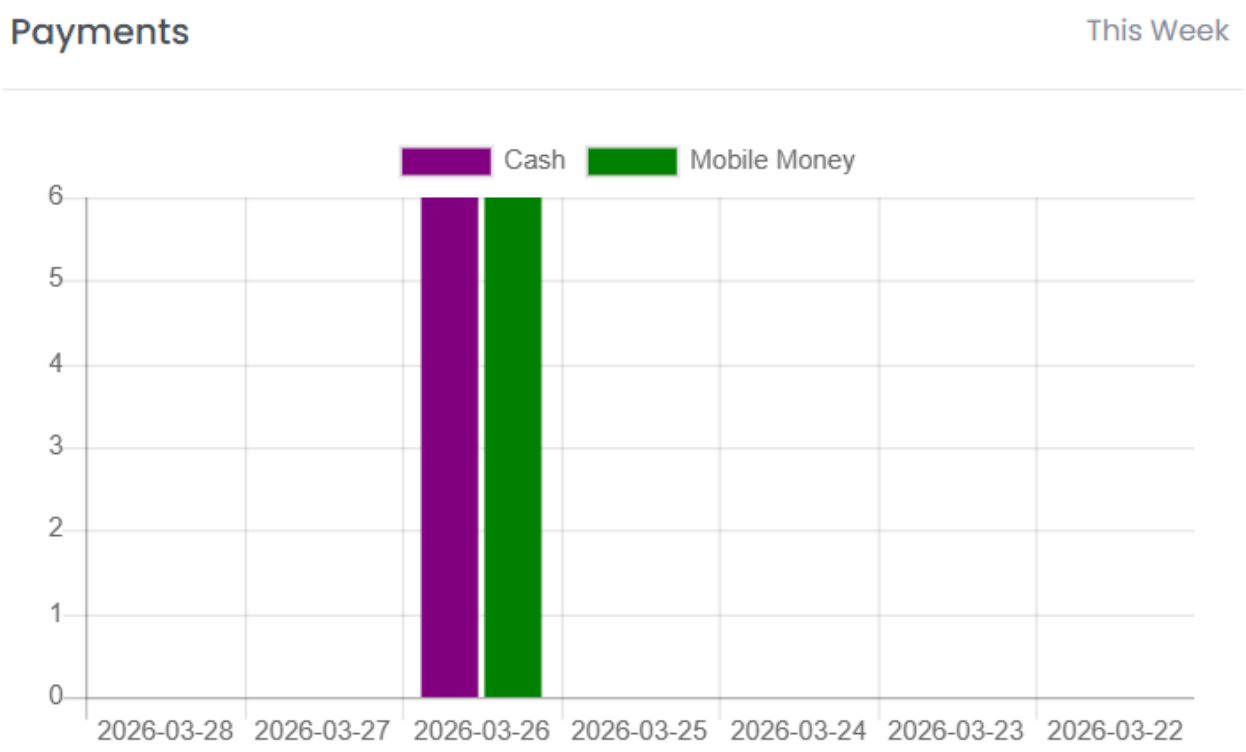


Stock Alerts

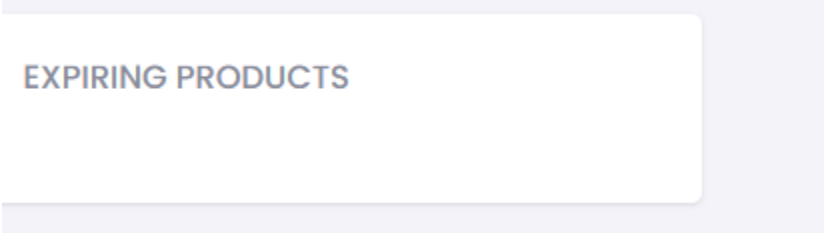
Product Name	Quantity	Supplier
PANADOL	10 Products	DAWA EAST AFRICA LIMITED
PAXLOVID	13 Products	KENTACKY PHARMACEUTICAL LTD
FLUCONAZOLE	34 Products	DAWA EAST AFRICA LIMITED

Sales overview shows total sales in a day in a chart that also shows total sales for each day, in the previous 7 days, and the pie chart shows the product that is sold in large numbers while the chart for payments shows which payment method was used by customer to pay for any sale made

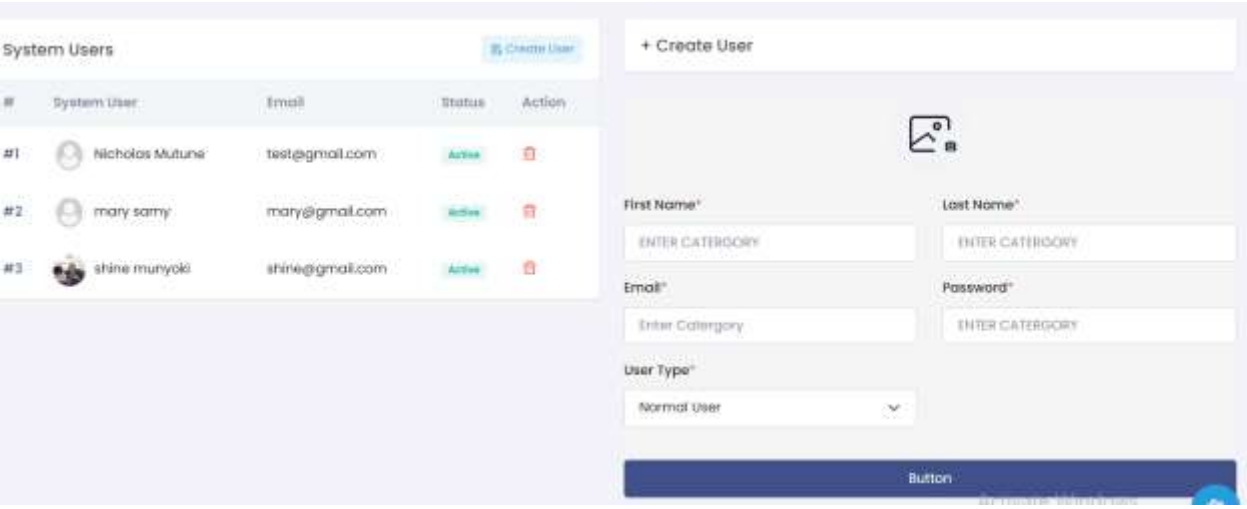




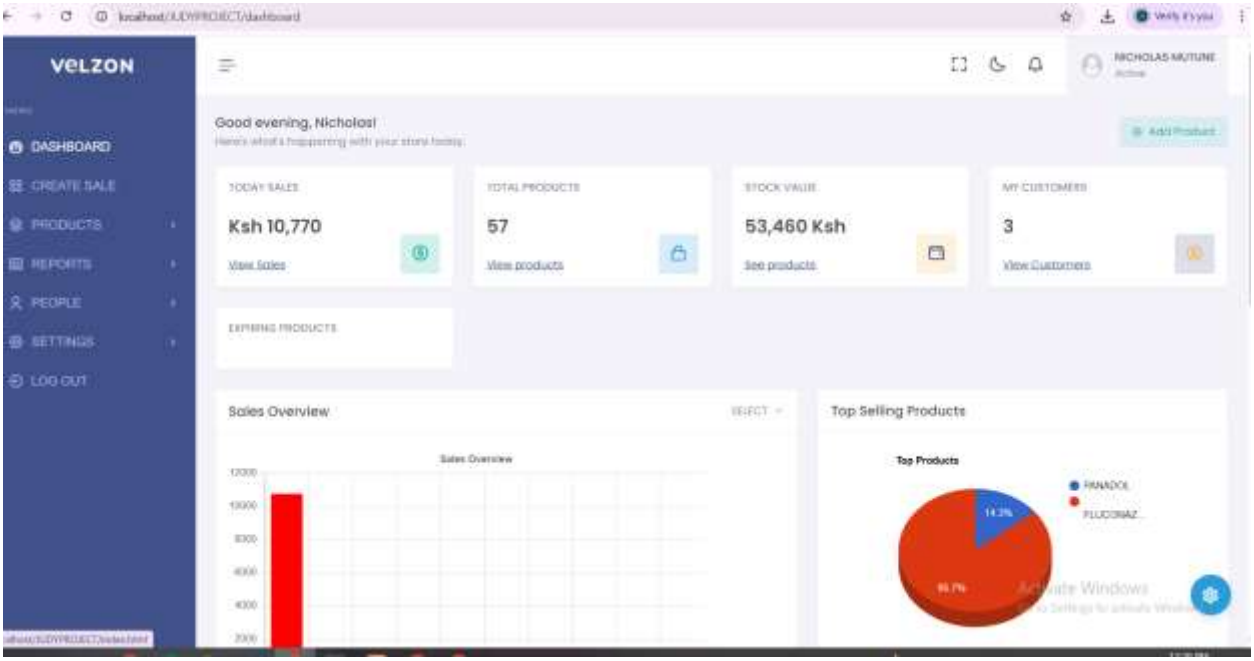
The products expiring in 5 months to come are displayed here, with a warning alert when logging in.

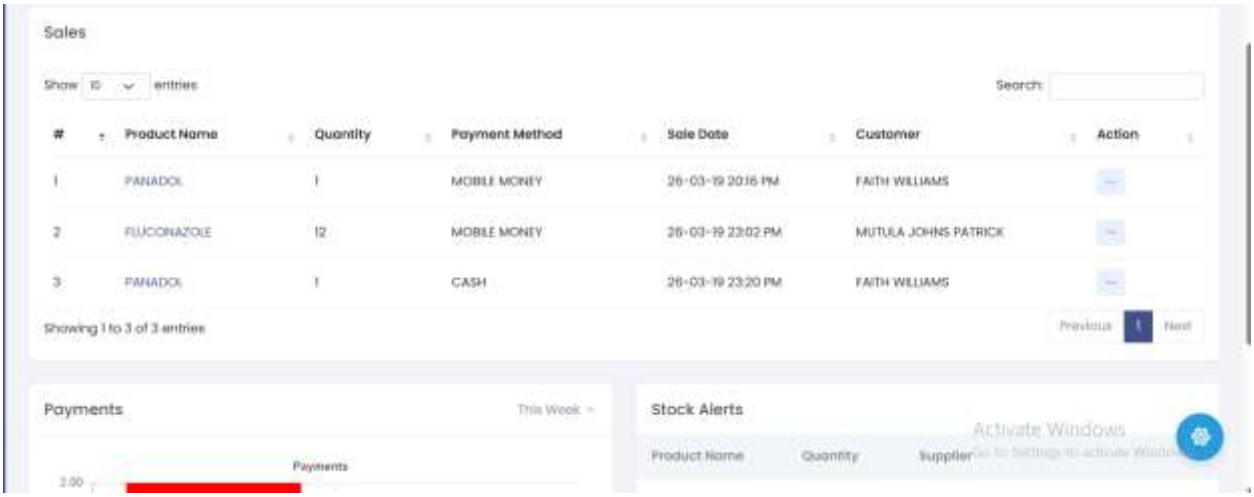


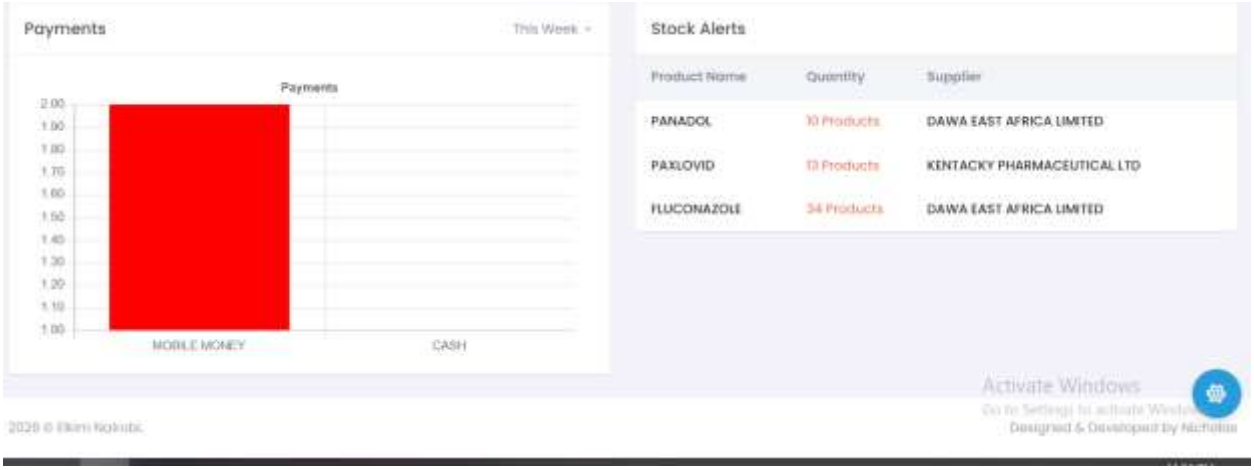
Eligible system users are managed here by the admin and one cannot change the login details unless authorized by the admin



Overall dashboard appearance below







Functional testing was performed on all major modules, including user authentication, product management, sales processing, customer management, supplier management, and category management. The login system was tested to ensure that only authorized users with valid email and password credentials can access the system. Invalid login attempts were correctly rejected, preventing unauthorized access.

Sales functionality was tested by selecting products, entering quantities, and verifying that the system correctly calculates total sales and profit. The system was also tested to confirm that sales are recorded accurately in the database and reflected correctly on the dashboard charts.

Validation testing was conducted both on the client side (JavaScript) and server side (PHP). Input fields were checked to ensure that required data is entered and that numeric fields

such as price and quantity contain valid values. This dual validation approach improves data accuracy and system reliability.

Integration testing was performed to ensure proper communication between system components. This included verifying that JavaScript correctly sends data to PHP scripts and that PHP correctly interacts with the MySQL database. The system responses were tested to ensure that appropriate notifications are displayed to the user.

Error testing was also carried out to identify and handle potential issues. For example, incorrect inputs, failed database operations, and missing data fields were tested. During testing, issues such as missing arrival and expiry dates in product records were identified, indicating the need to ensure proper data transmission and storage. These issues were addressed to improve system completeness.

The system was evaluated against the defined requirements, and the testing results confirmed that it performs its intended functions effectively, with accurate data processing, secure user authentication, and reliable system behavior.

4.5 Test Cases

Test Case ID	Functionality	Input	Expected Output	Actual Result	Status
TC01	User Login	Valid email & password	User is granted access to dashboard	Successful	Pass
TC02	User Login	Invalid email or password	Access denied with error message	Successful	Pass
TC03	Add Product	Complete product details	Product saved in database	Successful	Pass
TC04	Add Product	Missing required fields	System prevents submission or shows error	Successful	Pass
TC05	Record Sale	Select product + valid quantity	Total price and profit calculated correctly	Successful	Pass

TC06	Record Sale	Invalid quantity (e.g., text input)	Input rejected or error message displayed	Successful	Pass
TC07	Dashboard Charts	Existing sales data	Charts display correct sales trends and distribution	Successful	Pass
TC08	Add Customer	Valid customer details	Customer saved successfully	Successful	Pass
TC09	Add Supplier	Valid supplier details	Supplier saved successfully	Successful	Pass
TC10	Data Export	Export table to Excel	Excel file downloaded correctly	Successful	Pass
TC11	Product Dates Issue	Enter arrival & expiry dates	Dates saved in database	Initially Failed but later fixed Fixed	Pass

5 CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

The purpose of this project was to design and implement an Inventory and Sales Management System capable of improving the efficiency, accuracy, and reliability of business operations. Based on the analysis, design, and implementation carried out in the previous chapters, the system successfully meets its intended objectives. From the system analysis phase, it was established that manual or poorly structured systems lead to data inconsistencies, delays in processing transactions, and difficulties in tracking inventory and sales performance. The designed system addressed these challenges by providing a centralized database that integrates products, categories, suppliers, customers, users, and sales records into a single platform.

The design phase translated these requirements into a structured system architecture, including logical and physical designs, data flow diagrams, and an entity-relationship diagram. These models ensured that the system components were well organized and that data flows efficiently between different modules. The database design, in particular, supports data integrity through the

use of unique identifiers and relationships between tables. During implementation, the system was developed using modern web technologies including PHP, JavaScript, HTML, CSS, and MySQL within the XAMPP environment. The system supports key functionalities such as secure user authentication, product and inventory management, sales recording, and real-time data visualization through charts. The integration of frontend and backend components ensures smooth communication and efficient processing of user requests.

Testing results confirmed that the system performs as expected. Core functionalities such as login authentication, sales processing, and data management were validated successfully. The system also demonstrated reliability in handling user inputs and maintaining data accuracy through validation mechanisms. Overall, the developed system provides an effective solution for managing inventory and sales processes. It enhances operational efficiency, reduces human errors, and provides useful insights through data visualization. The system can therefore be considered a functional and reliable tool for supporting business decision-making. Although the system meets its core objectives, several improvements can be made to enhance its functionality and scalability in future versions.

First, the system can be improved by implementing stronger security measures such as password encryption and role-based access control. This will further protect sensitive user and business data from unauthorized access. Second, the system can be enhanced by improving the handling of date fields such as product arrival and expiry dates. These fields should be stored using appropriate date data types instead of text to allow better tracking, filtering, and reporting of expiring products. Third, the user interface can be further refined to improve user experience. This includes making the system more responsive for use on mobile devices and enhancing the visual presentation of dashboards and reports. Additionally, the system can be extended to

include advanced reporting features such as monthly and yearly sales analysis, profit trends, and automated report generation. This will provide deeper insights for business decision-making. Another important improvement is the integration of barcode scanning or automated product identification to speed up sales transactions and reduce manual input errors. Finally, the system can be deployed on a live server instead of a local environment to allow remote access and real-time usage by multiple users across different locations. In conclusion, while the current system provides a solid foundation for inventory and sales management, it also serves as a prototype for further development. Future enhancements can build upon this system to create a more robust, secure, and scalable solution.

6 References

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